



Sheet 7

The Bohr Atom

1. A hydrogen atom is in its ground state ($n = 1$). Using the Bohr theory of the atom, calculate (a) the radius of the orbit, (b) the linear momentum of the electron, (c) the angular momentum of the electron, (d) the kinetic energy, (e) the potential energy, and (f) the total energy.
2. The electron in a hydrogen atom at rest makes a transition from the $n = 2$ energy state to the $n = 1$ ground state. Find the wavelength, frequency, and energy of the emitted photon.
3. The Balmer series for the hydrogen atom corresponds to electronic transitions that terminate in the state of quantum number $n = 2$. (a) Find the longest-wavelength photon emitted and determine its energy. (b) Find the shortest-wavelength photon emitted in the Balmer series.
4. (a) What value of n is associated with the Lyman series line in hydrogen whose wavelength is 102.6 nm? (b) Could this wavelength be associated with the Paschen or Brackett series?
5. A hydrogen atom initially in its ground state ($n = 1$) absorbs a photon and ends up in the state for which $n = 3$. (a) What is the energy of the absorbed photon? (b) If the atom returns to the ground state, what photon energies could the atom emit?
6. Calculate the frequency of revolution and the orbit of the electron in the Bohr model of hydrogen for $n = 100$, 1000, and 10000. (b) Calculate the photon frequency for transitions from the n to $n - 1$ states for the same values of n as in part (a) and compare with the revolution frequencies found in part (a). (c) Explain how your results verify the correspondence principle.